MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCFTQ100 - BASIC BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Part A

Short Answer Questions

Answer any 5 questions

Each question carries 2 marks

1. **What are dextrins? [U]/[CO1]**
   Dextrins are low molecular weight carbohydrates produced by partial hydrolysis of starch
   They are water soluble and used as thickeners and binding agents in foods

2. **Discuss the importance of hydrogen bonding in stabilizing α helix. [U]/[CO2]**
   Hydrogen bonds form between peptide backbone groups
   They stabilize the helical structure of proteins and maintain secondary structure

3. **What is ketonic rancidity? [U]/[CO4]**
   Ketonic rancidity is spoilage caused by formation of ketones due to microbial action on fats
   It results in unpleasant odor and flavor in fatty foods

4. **What are aldopentoses? Give examples. [K]/[CO1]**
   Aldopentoses are five carbon monosaccharides containing an aldehyde group
   Examples: ribose, arabinose, xylose

5. **What are activators? Give examples. [U]/[CO3]**

   Activators are substances that increase enzyme activity

   Examples: Mg^{2+} , Zn^{2+} , Ca^{2+} ions

6. **What are trans fatty acids? How are they harmful to our body? [A]/[CO4]**

   Trans fatty acids are unsaturated fats with trans configuration of double bonds

   They increase LDL cholesterol, lower HDL cholesterol and raise cardiovascular disease risk

7. **Define Km. State its significance. [A]/[CO3]**

   Km (Michaelis constant) is substrate concentration at which enzyme activity reaches half of maximum velocity

   Indicates enzyme affinity for substrate; lower Km means higher affinity

Part B

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

8. **Prepare a note on the following reactions. [U]/[CO1]**

1) Reducing property of sugars

Reducing sugars possess free aldehyde or ketone groups

They reduce Benedict's or Fehling's reagents forming colored precipitates

Important in identification of carbohydrates

2) Dehydration reaction

Removal of water molecules from sugars by strong acids

Produces furfural or hydroxymethylfurfural derivatives

Basis of tests such as Molisch test

3) Osazone formation

Reducing sugars react with phenylhydrazine to form osazones

Characteristic crystals aid sugar identification

Example: glucose and fructose form similar osazones

4) Oxidation and reduction

Oxidation converts sugars into sugar acids

Reduction converts sugars into sugar alcohols

Important reactions in carbohydrate chemistry

9. **Discuss the impact of activators in an enzyme catalysed reaction. [E]/[CO3]**

Activators enhance enzyme activity and reaction rate

Often act as cofactors or metal ions required for catalysis

Help proper enzyme conformation and substrate binding

Examples include Mg^{2+} , Zn^{2+} and Ca^{2+}

Increase efficiency of biochemical reactions

10. **Differentiate between fats and oils. [An]/[CO4]**

Fats	**Oils**
Solid at room temperature	Liquid at room temperature
Rich in saturated fatty acids	Rich in unsaturated fatty acids
Mainly animal origin	Mainly plant origin
Higher melting point	Lower melting point
Example: butter	Example: sunflower oil

11. **Give the structural configuration of ketohexoses with empirical formula CH_2O_6 . [K]/[CO1]**

Ketohexoses are six carbon monosaccharides containing a ketone group

General formula: $C_6H_{12}O_6$

Ketone group usually at carbon 2

Examples: fructose, sorbose, tagatose

Exist in linear and cyclic forms

12. **Describe the effect of changing pH on altering the ionic state of amino acids. [A]/[CO2]**

Amino acids exist as zwitterions at isoelectric pH

At acidic pH they gain protons and become positively charged

At alkaline pH they lose protons and become negatively charged

pH changes affect solubility, structure and protein function

13. **What are lipids? Narrate the functions of lipids. [U]/[CO4]**

Lipids are hydrophobic organic compounds including fats, oils and related molecules

Function as major energy reserves

Structural components of cell membranes

Aid absorption of fat soluble vitamins

Provide insulation and protection

Serve as precursors of hormones and signaling molecules

14. **Explain the chemical properties of amino acids due to COOH group. [U]/[CO2]**

Carboxyl group imparts acidic nature to amino acids

Forms salts with bases

Undergoes esterification with alcohols

Participates in peptide bond formation

Can undergo decarboxylation to form amines

Part C

Long Essay Questions

Answer any 2 questions

Each question carries 10 marks

15. **Define active site. Outline the major salient features of active site. [K]/[CO3]**

Active site is the specific region of an enzyme where substrate binds and catalysis occurs

Possesses unique three dimensional structure complementary to substrate

Contains binding site and catalytic site

Highly specific for substrate recognition

Formed by specific amino acid residues

Provides suitable microenvironment for enzyme catalysis

Follows lock and key or induced fit mechanism

16. **Describe in detail on classification of carbohydrates. [U]/[CO1]**

Monosaccharides: simplest carbohydrates that cannot be hydrolyzed further

Classified as aldoses and ketoses

Based on carbon number: trioses, pentoses, hexoses

Examples: glucose, fructose, ribose

Oligosaccharides: contain 2–10 monosaccharide units

Disaccharides: sucrose, lactose, maltose

Trisaccharides and higher oligosaccharides also included

Polysaccharides: long chains of monosaccharides

Homopolysaccharides: starch, glycogen, cellulose

Heteropolysaccharides: pectin, hemicellulose

Storage and structural functions

17. **Write an essay on the structural organisation of proteins. Add a note on the chemical bonds responsible for its formation. [U]/[CO2]**

Primary structure- Linear sequence of amino acids linked by peptide bonds

Determines protein identity and function

Secondary structure- Includes α helix and β pleated sheet

Stabilized mainly by hydrogen bonds

Tertiary structure- Three dimensional folding of polypeptide chain

Determines biological activity

Stabilized by hydrogen bonds, ionic bonds, hydrophobic interactions and disulfide bonds

Quaternary structure- Association of two or more polypeptide subunits

Example: hemoglobin

Chemical bonds involved

Peptide bonds form primary structure

Hydrogen bonds stabilize secondary and tertiary structure

Disulfide bonds provide strength and stability

Ionic bonds and hydrophobic interactions aid folding and conformation

18. **Evaluate the structure, properties and functions of major phospholipids. [A]/[CO4]**

Phospholipids are amphipathic lipids containing glycerol, fatty acids, phosphate and nitrogenous base

Major phospholipids

Phosphatidylcholine (lecithin)

Phosphatidylethanolamine

Phosphatidylserine

Sphingomyelin

Properties

Amphipathic nature with hydrophilic and hydrophobic regions

Form bilayers in aqueous systems

Act as emulsifying agents

Functions

Major structural components of cell membranes

Involved in membrane fluidity and permeability

Function in lipid transport and signaling

Important in nerve tissue and lipoprotein structure